

FP

2. - Di Smith-Morra:

ip route add 10.28.40.0/24 via 10.28.41.226
ip route add 10.28.41.0/25 via 10.28.41.226
ip route add 10.28.41.128/26 via 10.28.41.226
ip route add 10.28.41.192/28 via 10.28.41.226
ip route add 10.28.41.208/29 via 10.28.41.226
ip route add 10.28.41.216/29 via 10.28.41.226

- Di Fianchetto:

ip route add 10.28.0.0/19 via 10.28.41.225
ip route add 10.28.32.0/21 via 10.28.41.225
ip route add 10.28.40.0/24 via 10.28.41.219
ip route add 10.28.41.0/25 via 10.28.41.219
ip route add 10.28.41.128/26 via 10.28.41.218
ip route add 10.28.41.192/28 via 10.28.41.219
ip route add 10.28.41.208/29 via 10.28.41.219

- Di Lucena:

ip route add 10.28.0.0/19 via 10.28.41.217
ip route add 10.28.32.0/21 via 10.28.41.217
ip route add 10.28.40.0/24 via 10.28.41.219
ip route add 10.28.41.0/25 via 10.28.41.219
ip route add 10.28.41.192/28 via 10.28.41.219
ip route add 10.28.41.208/29 via 10.28.41.219
ip route add 10.28.41.224/30 via 10.28.41.217

- Di Zwsichenzug:

ip route add 10.28.0.0/19 via 10.28.41.217
ip route add 10.28.32.0/21 via 10.28.41.217
ip route add 10.28.40.0/24 via 10.28.41.211
ip route add 10.28.41.0/25 via 10.28.41.211
ip route add 10.28.41.128/26 via 10.28.41.218
ip route add 10.28.41.192/28 via 10.28.41.211

```
ip route add 10.28.41.224/30 via 10.28.41.217
```

- Di Zugzwang:

```
ip route add 10.28.0.0/19 via 10.28.41.209
```

```
ip route add 10.28.32.0/21 via 10.28.41.209
```

```
ip route add 10.28.41.128/26 via 10.28.41.209
```

```
ip route add 10.28.41.216/29 via 10.28.41.209
```

```
ip route add 10.28.41.224/30 via 10.28.41.209
```

3. - Di DHCP Relay:

```
10.28.41.131 10.28.41.130
```

```
echo net.ipv4.ip_forward=1 > /etc/sysctl.conf
```

```
service isc-dhcp-relay restart
```

A. Smith-Morra:

```
eth1 eth2 eth3
```

B. Fianchetto, Lucena, Zwischenzug:

```
eth0 eth1
```

C. Zugzwang:

```
eth0 eth1 eth2 eth3
```

- Di DHCP Server (Master/Ponzianni):

A. nano /etc/default/isc-dhcp-server:

```
INTERFACESv4="eth0"
```

B. nano /etc/dhcp/dhcpd.conf:

```
failover peer "dhcp-failover" {  
    primary;          # Ponzianni server utama  
    address 10.28.41.130; # Alamat IP Ponzianni  
    port 647;         # Port komunikasi failover  
    peer address 10.28.41.131; # Alamat IP RuyLopez  
    peer port 647;     # Port komunikasi RuyLopez  
    max-response-delay 60;  
    max-unacked-updates 10;  
    mclt 3600;  
    split 128;
```

```
}
```

```
subnet 10.28.0.0 netmask 255.255.224.0 {  
    option routers 10.28.0.1;  
    option broadcast-address 10.28.31.255;  
    option domain-name-servers 10.28.41.194, 10.28.41.195, 192.168.122.1;  
    default-lease-time 600;  
    max-lease-time 7200;  
  
    pool {  
        failover peer "dhcp-failover";  
        range 10.28.0.10 10.28.31.254;  
    }  
}
```

```
subnet 10.28.32.0 netmask 255.255.248.0 {  
    option routers 10.28.32.1;  
    option broadcast-address 10.28.39.255;  
    option domain-name-servers 10.28.41.194, 10.28.41.195, 192.168.122.1;  
    default-lease-time 600;  
    max-lease-time 7200;  
  
    pool {  
        failover peer "dhcp-failover";  
        range 10.28.32.10 10.28.39.254;  
    }  
}
```

```
subnet 10.28.40.0 netmask 255.255.255.0 {  
    option routers 10.28.40.1;  
    option broadcast-address 10.28.40.255;
```

```
option domain-name-servers 10.28.41.194, 10.28.41.195, 192.168.122.1;

default-lease-time 600;

max-lease-time 7200;
```

```
pool {
    failover peer "dhcp-failover";
    range 10.28.40.10 10.28.40.254;
}
```

```
host Slav {
    hardware ethernet 02:42:04:eb:1c:00;
    fixed-address 10.28.40.2;
}
}
```

```
subnet 10.28.41.0 netmask 255.255.255.128 {
    option routers 10.28.41.1;
    option broadcast-address 10.28.41.127;
    option domain-name-servers 10.28.41.194, 10.28.41.195, 192.168.122.1;
    default-lease-time 600;
    max-lease-time 7200;
    pool {
        failover peer "dhcp-failover";
        range 10.28.41.10 10.28.41.126;
    }
}
```

```
host Sicillian {
    hardware ethernet 02:42:e1:03:0d:00;
    fixed-address 10.28.41.2;
}
}
```

```
subnet 10.28.41.128 netmask 255.255.255.192 {  
    option routers 10.28.41.129;  
    option broadcast-address 10.28.41.191;  
    option domain-name-servers 10.28.41.194, 10.28.41.195, 192.168.122.1;  
    default-lease-time 600;  
    max-lease-time 7200;  
    pool {  
        failover peer "dhcp-failover";  
        range 10.28.41.130 10.28.41.190;  
    }  
}
```

```
subnet 10.28.41.192 netmask 255.255.255.240 {  
    option routers 10.28.41.193;  
    option broadcast-address 10.28.41.207;  
    option domain-name-servers 10.28.41.194, 10.28.41.195, 192.168.122.1;  
    default-lease-time 600;  
    max-lease-time 7200;  
    pool {  
        failover peer "dhcp-failover";  
        range 10.28.41.194 10.28.41.206;  
    }  
}
```

C. service isc-dhcp-server restart

D. service isc-dhcp-server status

- Di DHCP Server (Slave/RuyLopez):

A. nano /etc/default/isc-dhcp-server:

```
INTERFACESv4="eth0"
```

B. nano /etc/dhcp/dhcpd.conf:

```
failover peer "dhcp-failover" {
```

```
secondary;          # RuyLopez server secondary
address 10.28.41.131; # Alamat IP RuyLopez
port 647;           # Port komunikasi failover
peer address 10.28.41.130; # Alamat IP Ponziani
peer port 647;      # Port komunikasi Ponziani
}
```

```
subnet 10.28.0.0 netmask 255.255.224.0 {
    option routers 10.28.0.1;
    option broadcast-address 10.28.31.255;
    option domain-name-servers 192.168.122.1;
    default-lease-time 600;
    max-lease-time 7200;

    pool {
        failover peer "dhcp-failover";
        range 10.28.0.10 10.28.0.20;
    }
}
```

```
subnet 10.28.32.0 netmask 255.255.248.0 {
    option routers 10.28.32.1;
    option broadcast-address 10.28.39.255;
    option domain-name-servers 192.168.122.1;
    default-lease-time 600;
    max-lease-time 7200;

    pool {
        failover peer "dhcp-failover";
        range 10.28.32.10 10.28.32.20;
    }
}
```

```
}
```

```
subnet 10.28.40.0 netmask 255.255.255.0 {  
    option routers 10.28.40.1;  
    option broadcast-address 10.28.40.255;  
    option domain-name-servers 10.28.41.194, 10.28.41.195, 192.168.122.1;  
    default-lease-time 600;  
    max-lease-time 7200;
```

```
    pool {  
        failover peer "dhcp-failover";  
        range 10.28.40.10 10.28.40.254;  
    }
```

```
    host Slav {  
        hardware ethernet 02:42:04:eb:1c:00;  
        fixed-address 10.28.40.2;  
    }
```

```
}
```

```
subnet 10.28.41.0 netmask 255.255.255.128 {  
    option routers 10.28.41.1;  
    option broadcast-address 10.28.41.127;  
    option domain-name-servers 10.28.41.194, 10.28.41.195, 192.168.122.1;  
    default-lease-time 600;  
    max-lease-time 7200;  
    pool {  
        failover peer "dhcp-failover";  
        range 10.28.41.10 10.28.41.126;  
    }
```

```
host Sicilian {  
    hardware ethernet 02:42:e1:03:0d:00;  
    fixed-address 10.28.41.2;  
}  
}
```

```
subnet 10.28.41.128 netmask 255.255.255.192 {  
    option routers 10.28.41.129;  
    option broadcast-address 10.28.41.191;  
    option domain-name-servers 10.28.41.194, 10.28.41.195, 192.168.122.1;  
    default-lease-time 600;  
    max-lease-time 7200;  
    pool {  
        failover peer "dhcp-failover";  
        range 10.28.41.130 10.28.41.190;  
    }  
}
```

```
subnet 10.28.41.192 netmask 255.255.255.240 {  
    option routers 10.28.41.193;  
    option broadcast-address 10.28.41.207;  
    option domain-name-servers 10.28.41.194, 10.28.41.195, 192.168.122.1;  
    default-lease-time 600;  
    max-lease-time 7200;  
    pool {  
        failover peer "dhcp-failover";  
        range 10.28.41.194 10.28.41.206;  
    }  
}
```

C. service isc-dhcp-server restart

D. service isc-dhcp-server status

5. - Di Caro-Kann (DNS Master):

A. nano /myscripts/dns/named.conf:

```
options {  
    directory "/myscripts/dns";  
    listen-on { any; };  
    allow-query { any; };  
    allow-transfer { 10.28.41.195; }; # transfer ke slave (Alekhine)  
    dnssec-validation no;  
};
```

```
zone "parkov.com" {  
    type master;  
    file "db.parkov";  
};
```

```
zone "paskarov.com" {  
    type master;  
    file "db.paskarov";  
};
```

```
zone "openings.com" {  
    type master;  
    file "db.openings";  
};
```

B. nano /myscripts/dns/db.parkov:

\$TTL 86400

```
@ IN SOA ns1.parkov.com. admin.parkov.com. (  
    1          ; serial  
    3600      ; refresh  
    1800      ; retry  
    604800    ; expire
```

```
      86400      ; minimum
)
```

```
@ IN NS ns1.parkov.com.
ns1 IN A 10.28.41.194
```

```
@ IN NS ns2.parkov.com.
ns2 IN A 10.28.41.195
```

```
@ IN A 10.28.40.2
www IN CNAME parkov.com.
C. nano /myscripts/dns/db.paskarov:
$TTL 86400
```

```
@ IN SOA ns1.paskarov.com. admin.paskarov.com. (
    1          ; serial
    3600       ; refresh
    1800       ; retry
    604800     ; expire
    86400      ; minimum
)
```

```
@ IN NS ns1.paskarov.com.
ns1 IN A 10.28.41.194
```

```
@ IN NS ns2.paskarov.com.
ns2 IN A 10.28.41.195
```

```
@ IN A 10.28.41.2
www IN CNAME paskarov.com.
D. nano /myscripts/start.sh:
#!/bin/sh
```

```
named -g -c /myscripts/dns/named.conf
```

- Di Alekhine (DNS Slave):

A. nano /myscripts/dns/named.conf:

```
options {  
    directory "/myscripts/dns";  
    listen-on { any; };  
    allow-query { any; };  
    dnssec-validation no;  
};
```

```
zone "parkov.com" {  
    type slave;  
    file "db.parkov";  
    masters { 10.28.41.194; };  
};
```

```
zone "paskarov.com" {  
    type slave;  
    file "db.paskarov";  
    masters { 10.28.41.194; };  
};
```

```
zone "openings.com" {  
    type slave;  
    file "db.openings";  
    masters { 10.28.41.194; };  
};
```

B. nano /myscripts/start.sh:

```
#!/bin/sh
```

```
named -g -c /myscripts/dns/named.conf
```

7. - Di Sicilian:

A. nano /myscripts/myweb/index.html: [lni](#)

B. nano /myscripts/myconfig/nginx.conf:

```
user www-data;
```

```
worker_processes auto;
```

```
worker_cpu_affinity auto;
```

```
pid /tmp/nginx.pid;
```

```
error_log /myscripts/mylogs/error.log;
```

```
events { worker_connections 768; }
```

```
http {
```

```
    log_format no7 '[$time_local] Jarkom Node Sicilian Access from $remote_addr using  
method "$request" returned status $status with $body_bytes_sent bytes sent in $request_time  
seconds';
```

```
    server {
```

```
        listen 80;
```

```
        server_name _;
```

```
        access_log /tmp/access.log no7;
```

```
        root /myscripts/myweb;
```

```
        index index.html;
```

```
        location / {
```

```
            try_files $uri $uri/ =404;
```

```
        }
```

```
    }
```

```
}
```

C. nano /myscripts/myconfig/start_http.sh:

```
#!/bin/bash
```

```
nginx -c /myscripts/myconfig/nginx.conf -g 'daemon off;'
```

9. – Di Slav:

A. nano /myscripts/myweb/index.html: [lni](#)

B. nano /myscripts/myconfig/nginx.conf:

```
user www-data;
```

```
worker_processes auto;
```

```
worker_cpu_affinity auto;
```

```
pid /tmp/nginx.pid;
```

```
error_log /myscripts/mylogs/error.log;
```

```
events { worker_connections 768; }
```

```
http {
```

```
    server {
```

```
        listen 8000;
```

```
        listen 8888;
```

```
        server_name _;
```

```
        root /myscripts/myweb;
```

```
        index index.html;
```

```
        location / {
```

```
            try_files $uri $uri/ =404;
```

```
        }
```

```
    }
```

```
}
```

C. nano /myscripts/myconfig/start_http.sh:

```
#!/bin/bash
```

```
nginx -c /myscripts/myconfig/nginx.conf -g 'daemon off;'
```

10. - Di Caro-Kann:

A. nano /myscripts/dns/db.openings:

```
$TTL 86400
```

```
@ IN SOA ns1.openings.com. admin.openings.com. (  
    1          ; serial  
    3600      ; refresh  
    1800      ; retry  
    604800    ; expire  
    86400     ; minimum  
)
```

```
@ IN NS ns1.openings.com.  
ns1 IN A 10.28.41.194
```

```
@ IN NS ns2.openings.com.  
ns2 IN A 10.28.41.195
```

```
@ IN A 10.28.41.210  
www IN CNAME openings.com.  
B. sh /myscripts/start.sh:
```

- Di Petrov:

A. nano /myscripts/myconfig/nginx.conf:

```
user www-data;  
  
worker_processes auto;  
  
worker_cpu_affinity auto;  
  
pid /tmp/nginx.pid;  
  
error_log /myscripts/mylogs/error.log;
```

```
events { worker_connections 768; }
```

```
http {  
    upstream roundrobin {  
        server 10.28.41.2;  
        server 10.28.40.2:8000;
```

```

server 10.28.40.2:8888;
}

server {
    listen 80;
    server_name _;

    location = / {
        proxy_pass http://roundrobin/;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }

    location = /slav {
        proxy_pass http://10.28.40.2:8000/;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }

    location = /sicilian {
        proxy_pass http://10.28.41.2/;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }

    location / {
        rewrite ^ / last;
    }
}

```

B. nano /myscripts/myconfig/start_http.sh:

```
#!/bin/bash
```

```
nginx -c /myscripts/myconfig/nginx.conf -g 'daemon off;'
```

12. - Di Semua Router:

```
cd /usr/lib/frr
```

```
./zebra -d
```

```
./ripd -d
```

```
./mgmtd -d
```

```
vttysh
```

```
conf t
```

```
router rip
```

- Smith-Morra:

```
network 10.28.0.0/19
```

```
network 10.28.32.0/21
```

```
network 10.28.41.224/30
```

```
default-information originate
```

- Fianchetto:

```
network 10.28.41.216/29
```

```
network 10.28.41.224/30
```

- Lucena:

```
network 10.28.41.128/26
```

```
network 10.28.41.216/29
```

- Zwischenzug:

```
network 10.28.41.208/29
```

```
network 10.28.41.216/29
```

- Zugzwang:

```
network 10.28.40.0/24
```

```
network 10.28.41.0/25
```

```
network 10.28.41.192/28
```

```
network 10.28.41.208/29
```

- Di Semua Router:

```
exit, show ip rip, show ip route
```

13. - Di Smith-Morra:

```
iptables -A FORWARD -s 10.28.0.0/19 -d 10.28.41.192/28 -p tcp --dport 22 -j DROP
```

```
iptables -A FORWARD -s 10.28.32.0/21 -d 10.28.41.192/28 -p tcp --dport 22 -j DROP
```

```
iptables-save
```

- Tes di Carro-Kann & Alekhine:

```
nc -l -p 22
```

- Tes di Stafford & Blackmar-Diemer:

```
nc 10.28.41.194 22
```

```
nc 10.28.41.195 22
```

14. - Di masing-masing Web Server (Slav dan Sicilian):

```
iptables -A INPUT -s 10.28.0.0/19 -m time --timestart 09:00 --timestop 17:00 --weekdays  
Mon,Tue,Wed,Thu,Fri -j ACCEPT
```

```
iptables -A INPUT -s 10.28.32.0/21 -m time --timestart 09:00 --timestop 17:00 --weekdays  
Mon,Tue,Wed,Thu,Fri -j ACCEPT
```

```
iptables -A INPUT -s 10.28.0.0/19 -j DROP
```

```
iptables -A INPUT -s 10.28.32.0/21 -j DROP
```

```
iptables-save
```

- Tes di Client-Group-3:

A. Benar: `date -s "2025-11-21 10:00:00"`

B. Salah: `date -s "2025-11-21 20:00:00"`

C. `telnet 10.28.40.2`

D. `telnet 10.28.41.2`

15. - Di Smith-Morra:

```
iptables -F
```

```
iptables -A FORWARD -s 10.28.0.0/19 -d 10.28.41.192/28 -p tcp --dport 22 -j NFLOG --nflog-prefix  
"DROP-SSH-019: " --nflog-group 0
```

```
iptables -A FORWARD -s 10.28.32.0/21 -d 10.28.41.192/28 -p tcp --dport 22 -j NFLOG --nflog-prefix  
"DROP-SSH-021: " --nflog-group 0
```

```
iptables -A FORWARD -s 10.28.0.0/19 -d 10.28.41.192/28 -p tcp --dport 22 -j DROP
```

```
iptables -A FORWARD -s 10.28.32.0/21 -d 10.28.41.192/28 -p tcp --dport 22 -j DROP
```

```
iptables -A INPUT -m conntrack --ctstate INVALID -j NFLOG --nflog-prefix "DROP INVALID IN: " --nflog-  
group 0
```

```
iptables -A FORWARD -m conntrack --ctstate INVALID -j NFLOG --nflog-prefix "DROP INVALID FW: " --nflog-group 0
```

```
iptables -A OUTPUT -m conntrack --ctstate INVALID -j NFLOG --nflog-prefix "DROP INVALID OUT: " --nflog-group 0
```

```
iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE -s 10.28.0.0/16
```

```
iptables-save
```

- Tes di Stafford:

```
ssh 10.28.41.194
```

```
ssh 10.28.41.195
```

- Cek log di Smith-Morra:

```
tail -F /var/log/ulog/syslogemu.log
```